Multiscale quantitative fractography of soft cut surfaces

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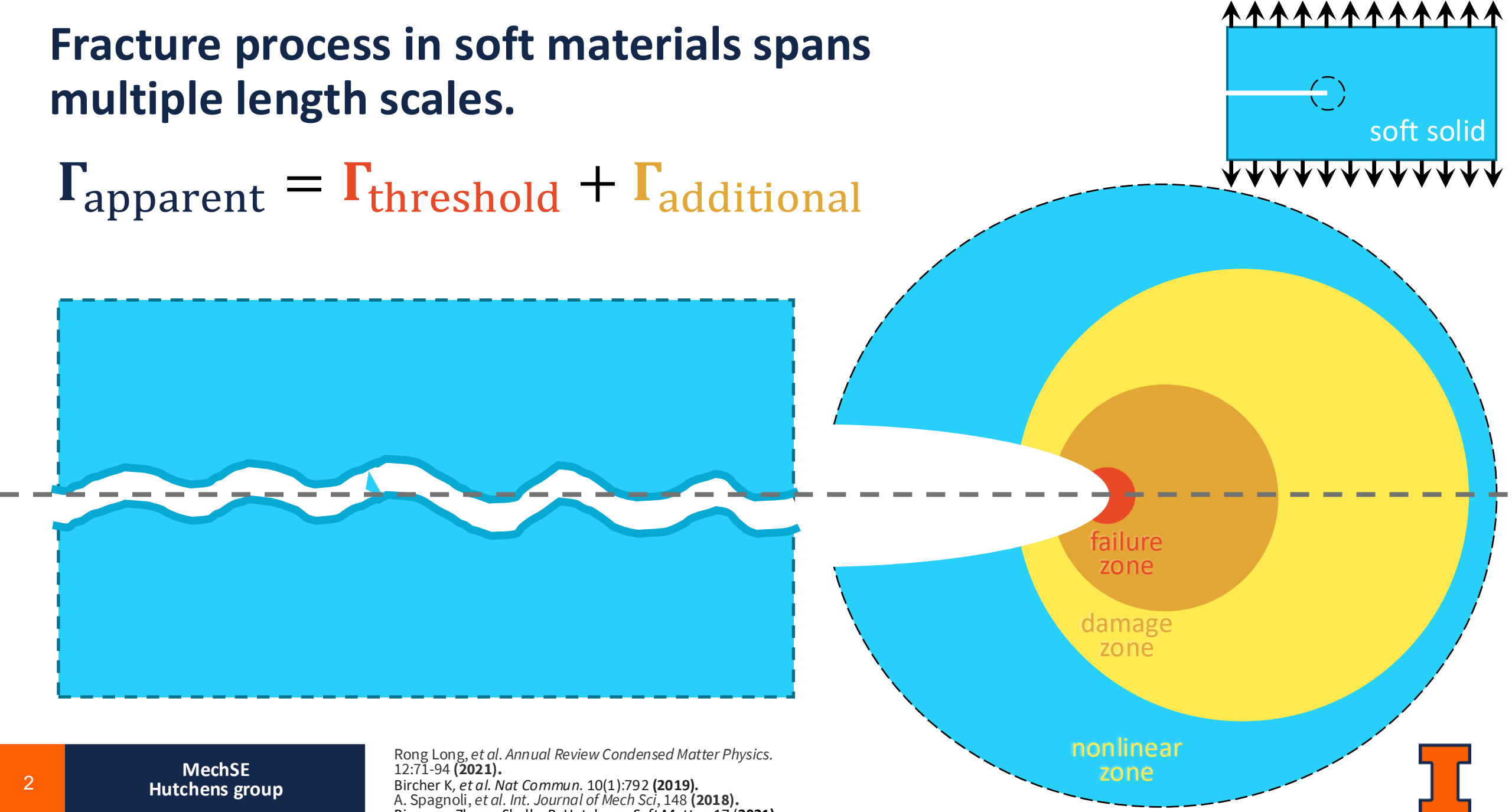
Peebles talk
Adhesion Society 2026

ILLINOIS

Fracture process in soft materials spans multiple length scales.

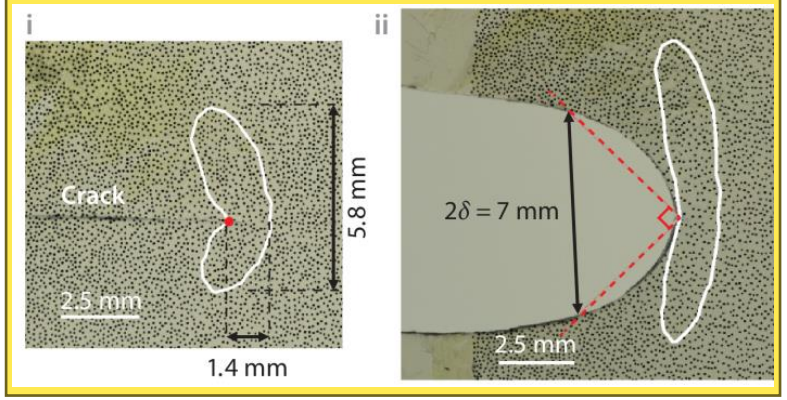
$$\Gamma_{\text{apparent}} = \Gamma_{\text{threshold}} + \Gamma_{\text{additional}}$$



Rong Long, *et al. Annual Review Condensed Matter Physics.* 12:71-94 **(2021).**
Bircher K, *et al. Nat Commun.* 10(1):792 **(2019).**
A. Spagnoli, *et al. Int. Journal of Mech Sci*, 148 **(2018).**
Bingyang Zhang, Shelby B. Hutchens. *Soft Matter,* 17 **(2021).**

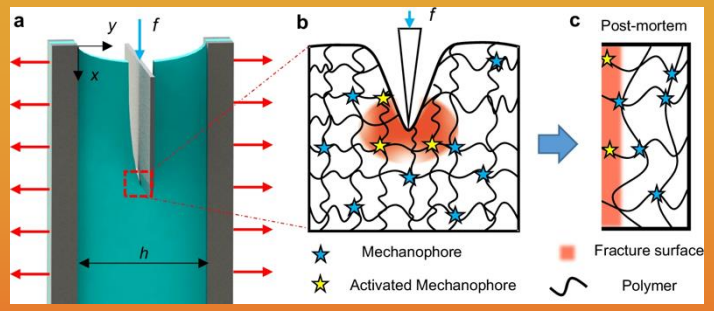
Full-field methods

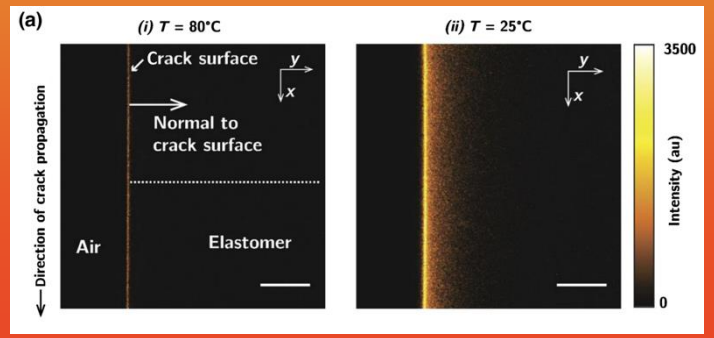
Rong Long, *et al. Annual Review Condensed Matter Physics*. 12:71-94 **(2021).**



Mechanophores

Zhao, D. *et al. Nat Commun* 16, 3203 **(2025).**
Slootman, J. *et al. Phys. Rev. X* 10, 041045 **(2020).**



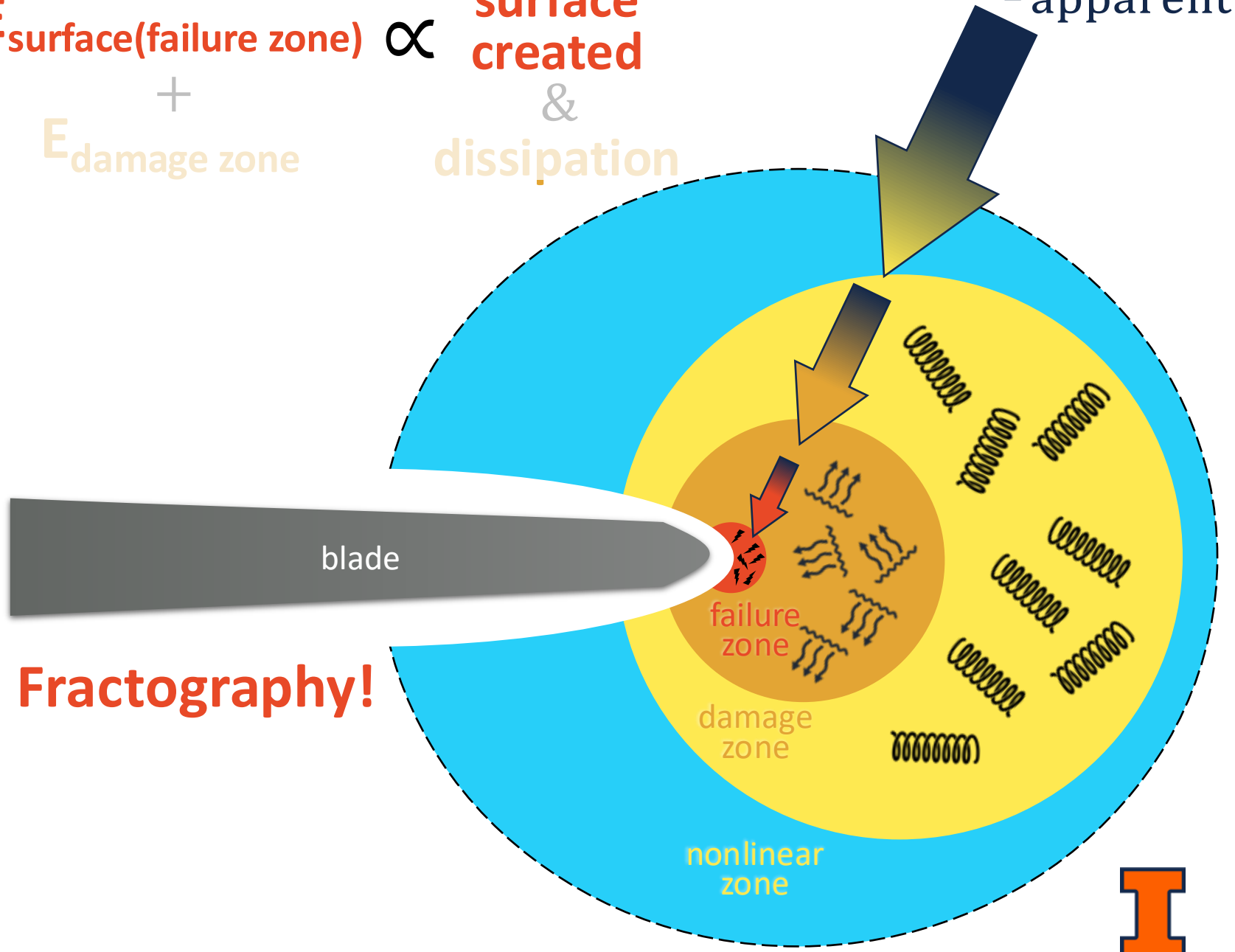


$E_{surface(failure\ zone)} \propto$ surface created

$+$ & dissipation

$E_{damage\ zone}$

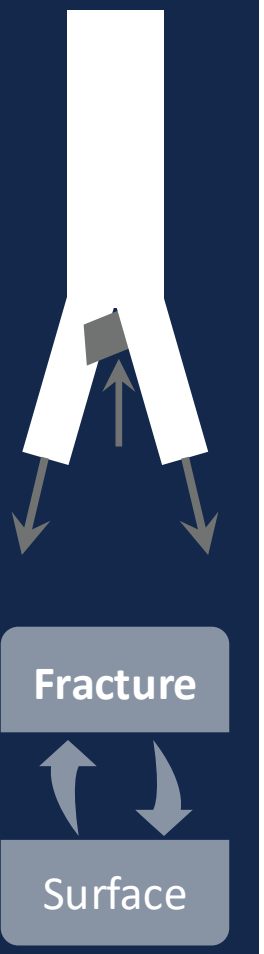
$\Gamma_{apparent}$

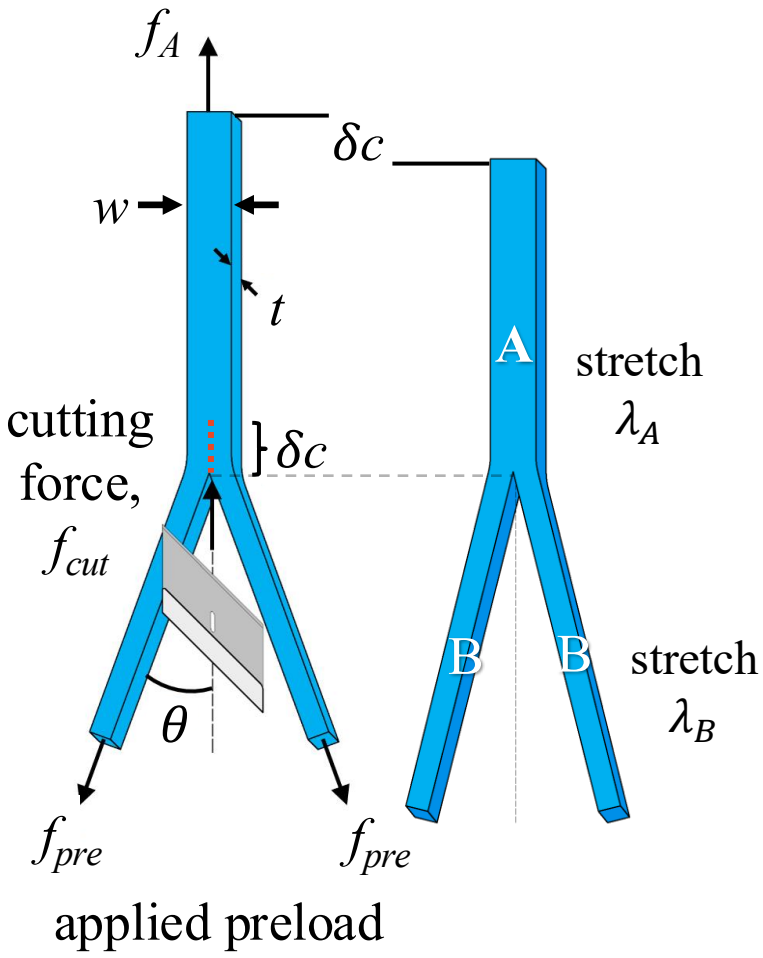


Fractography!

MechSE
Hutchens group

Controlled Y-shaped cutting allows variation in apparent fracture energy.

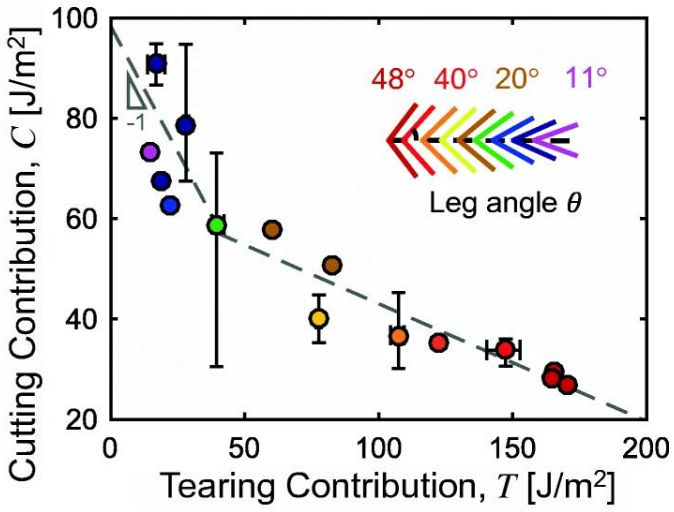


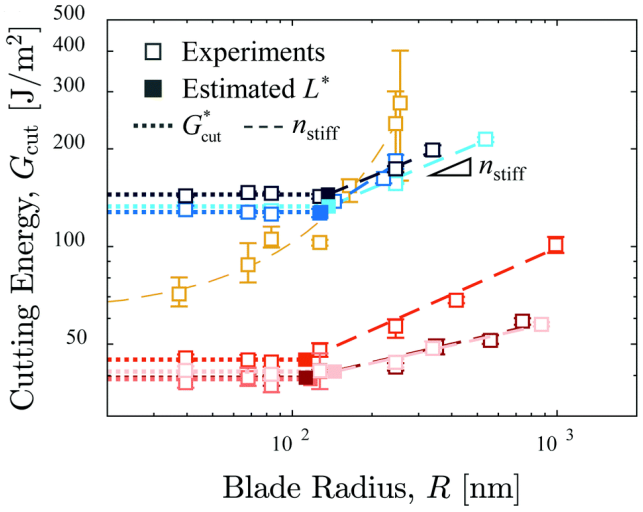


Cutting Energy Release Rate

linear approximation

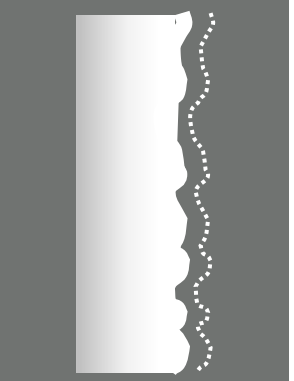
$$G_{cut} \approx \underbrace{\frac{f_{pre}\bar{\lambda}}{t}(1-\cos\theta)}_{\text{applied tearing, } T} + \underbrace{\frac{f_{cut}\bar{\lambda}}{2t}}_{\text{measured cutting, } C}$$





Lake & Yeoh. Int J. Fract 14. ***1978***
Bingyang Zhang, Shelby B. Hutchens. Soft Matter, 17. ***2021***

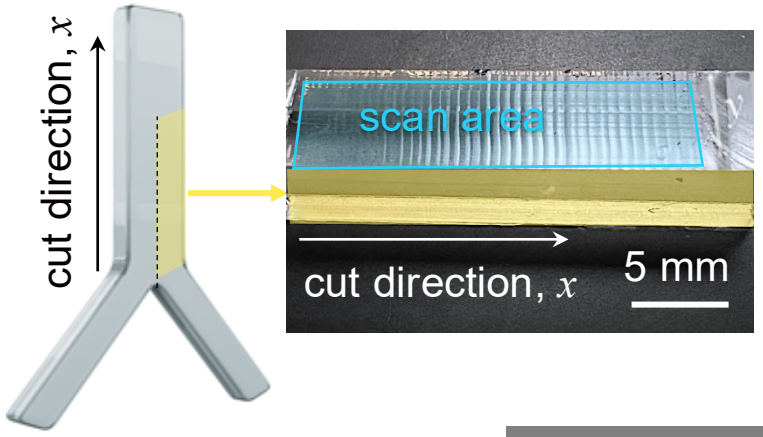
There are micro and milli-scale features over ~2 cm long surface



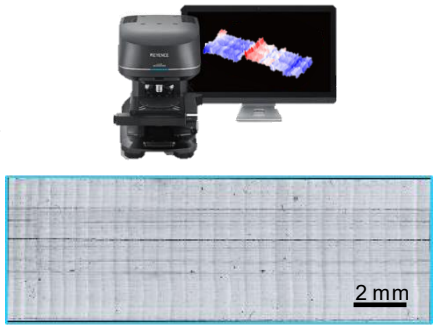
Fracture

Surface

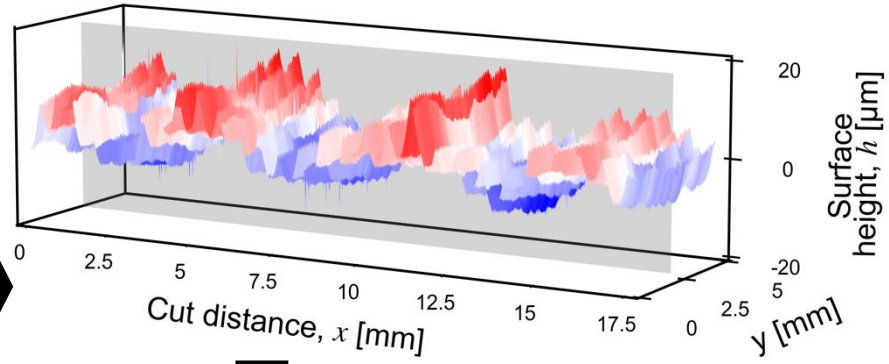
Cut surface generation
Y-shaped cutting



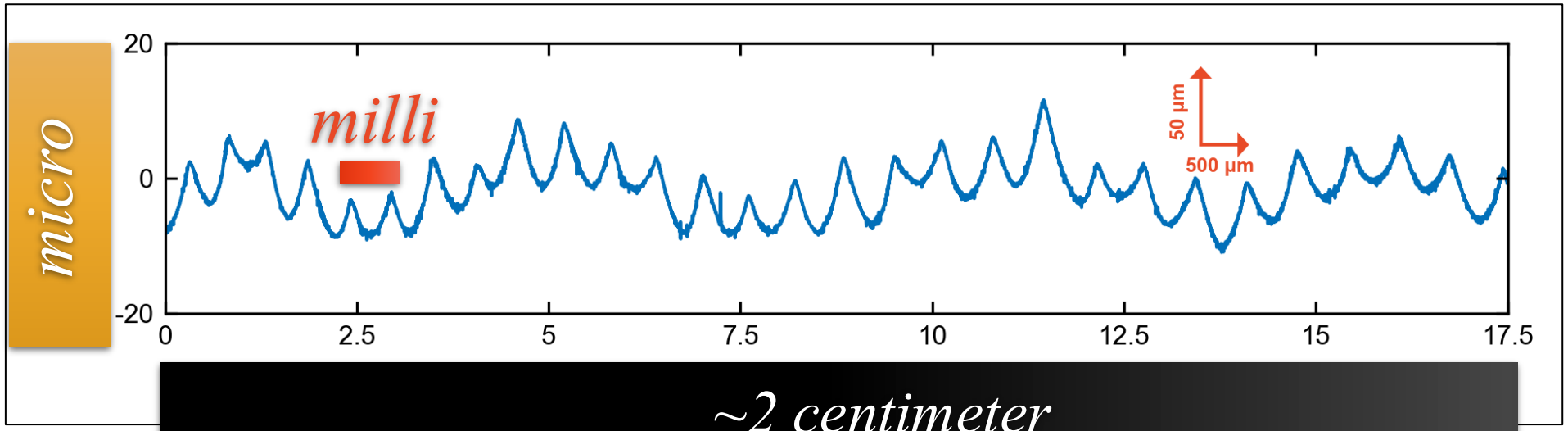
Optical topography
Height data over region of interest



Flattened height data

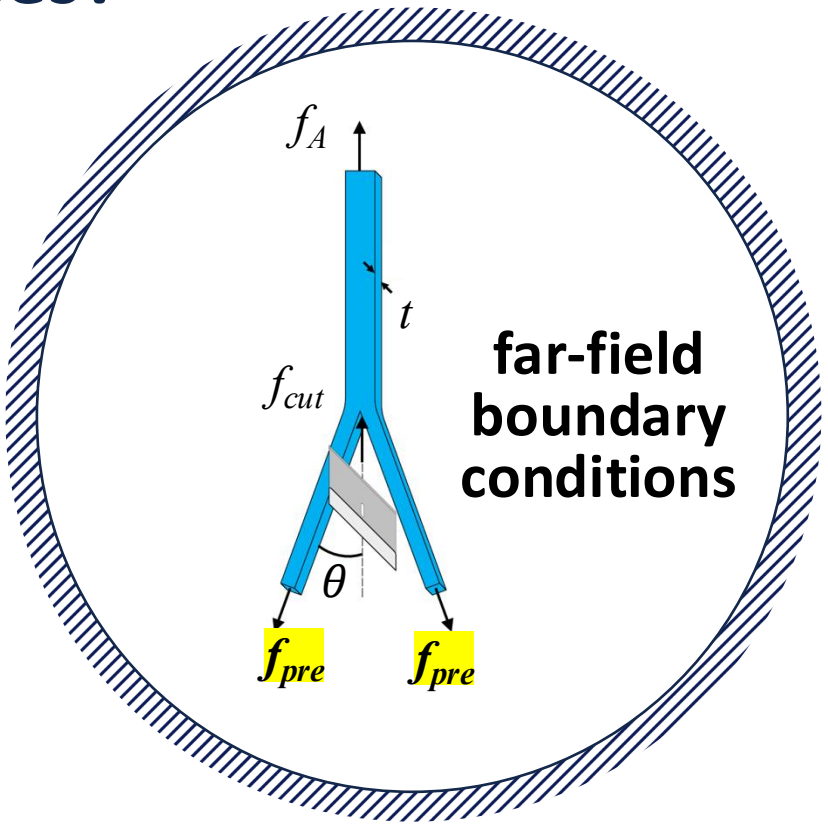


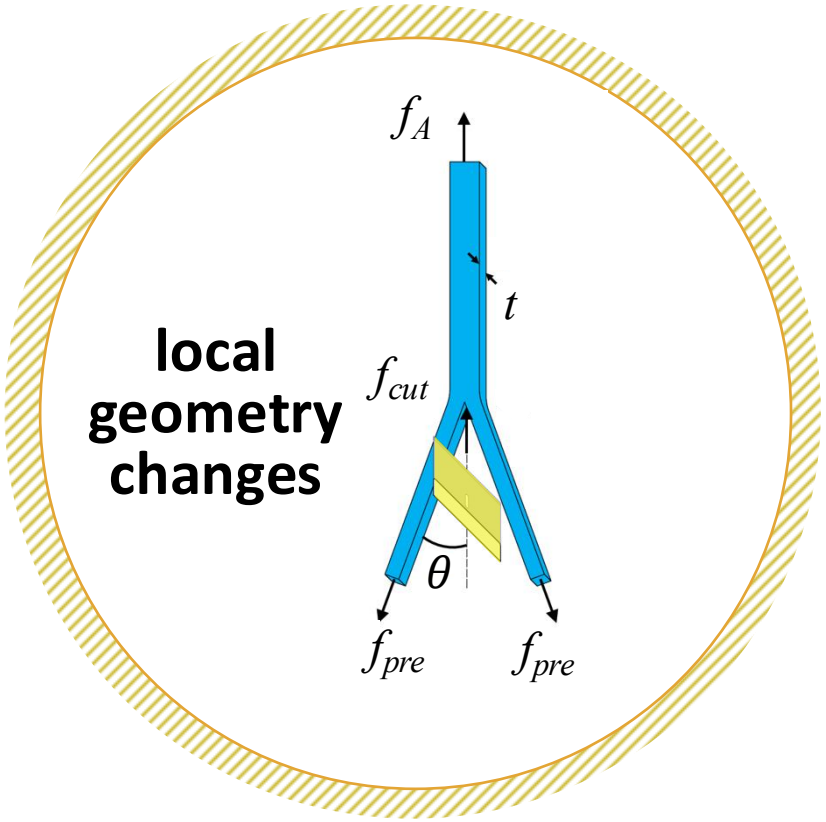
Line scan along cutting (x) direction

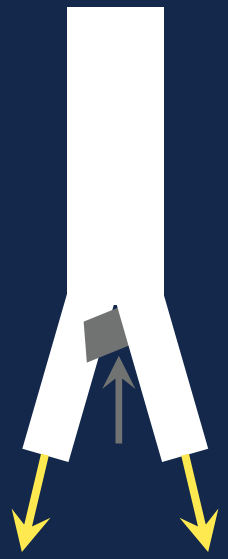


~2 centimeter

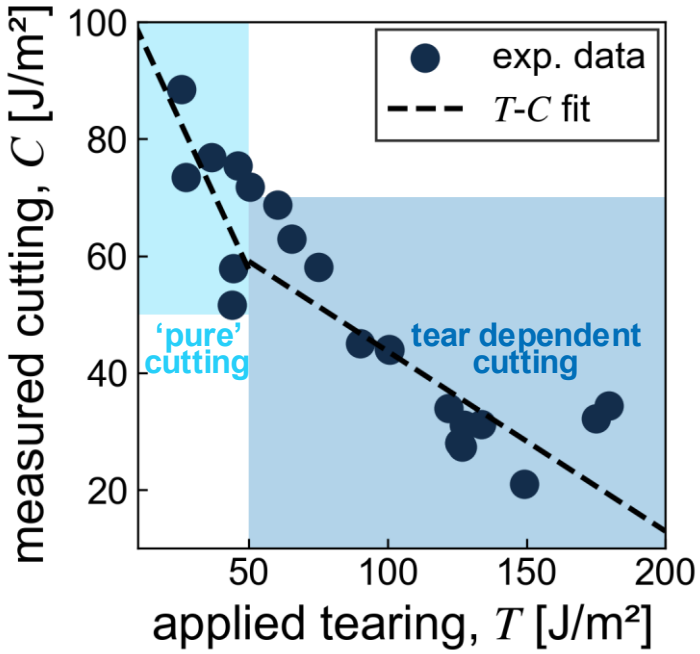
Is the apparent fracture energy reflected in the resulting cut surfaces?

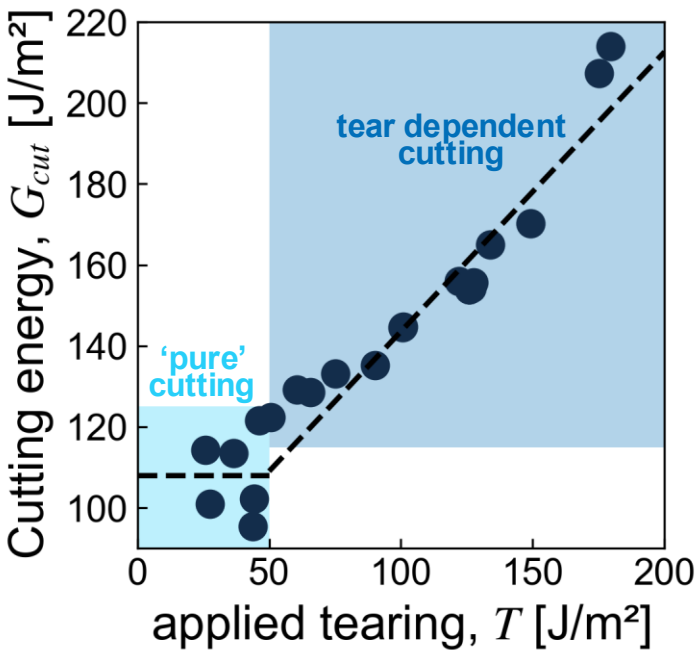


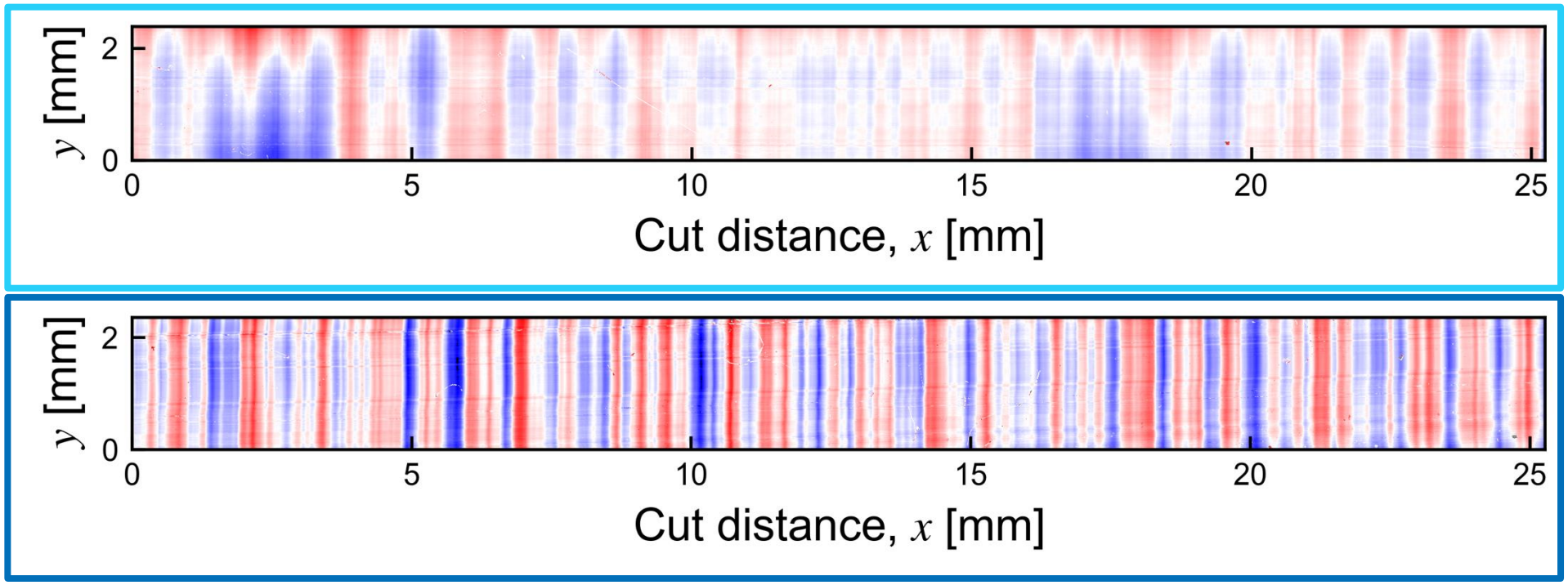




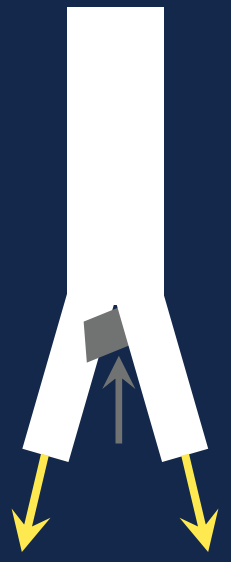
far-field boundary conditions



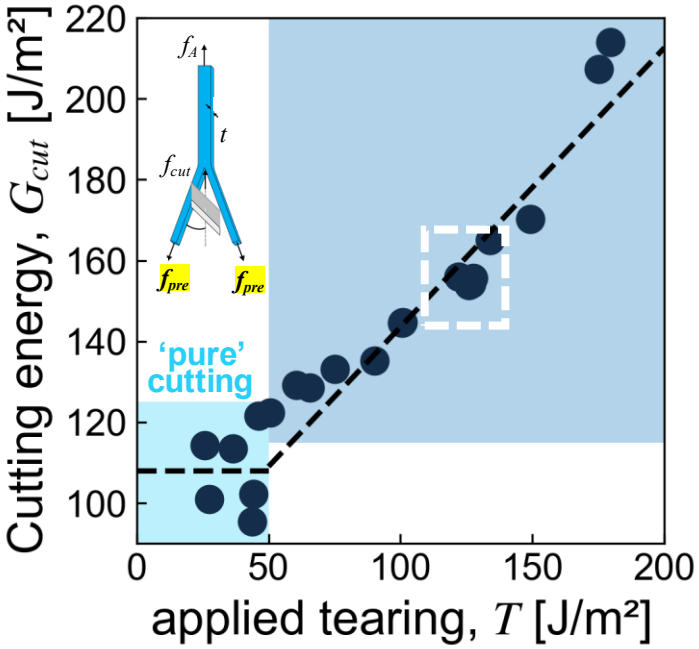


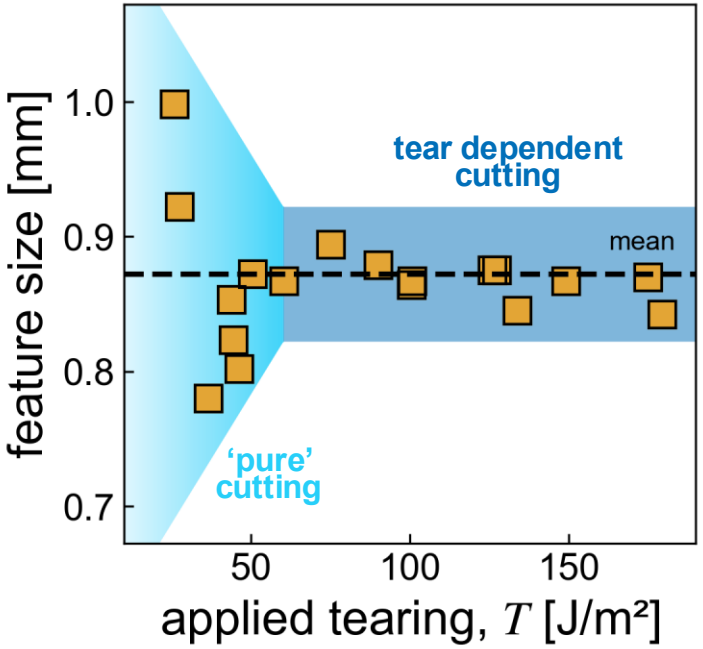


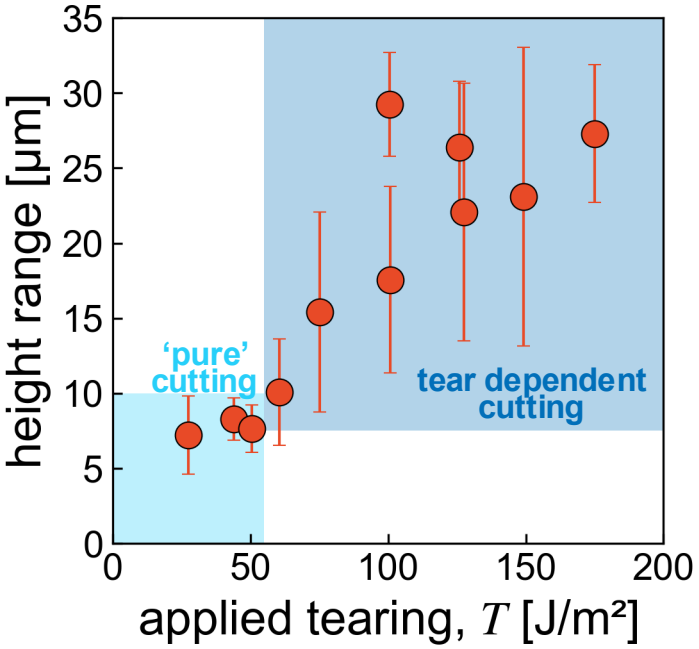
Increasing in far field applied tearing creates more prominent surface features.



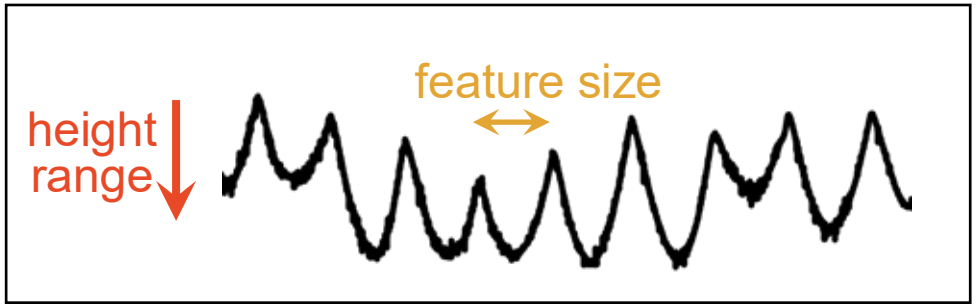
far-field boundary conditions



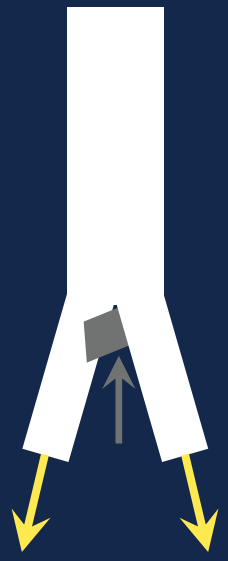




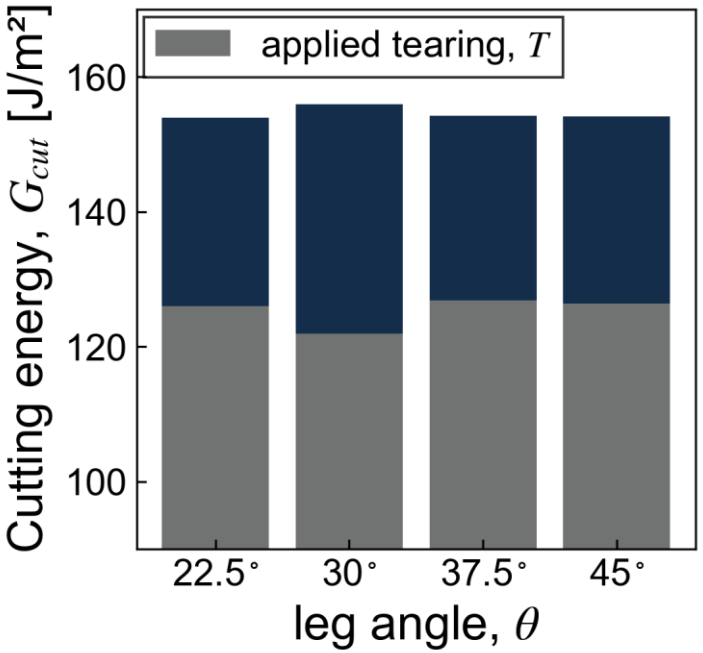
$$G_{cut} = T + C$$

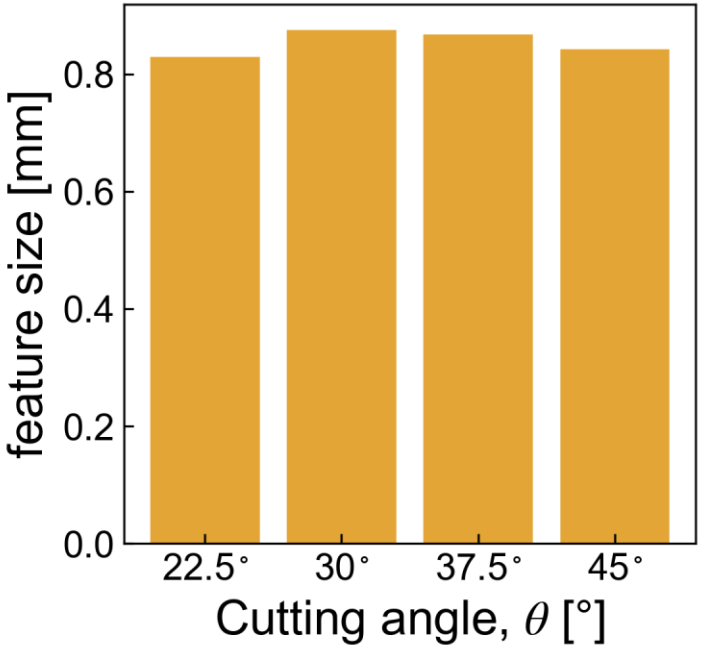


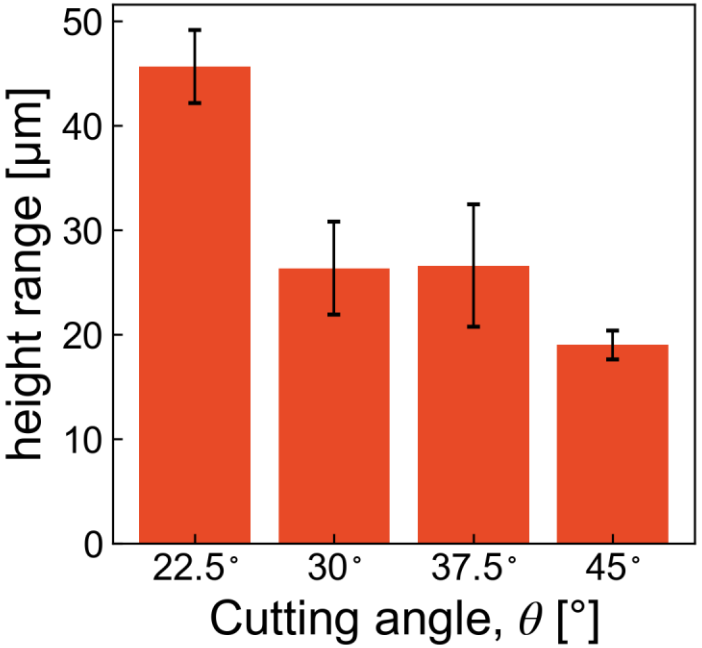
Surface characteristics also depend on the local stress distribution.

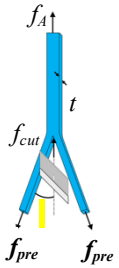


far-field boundary conditions



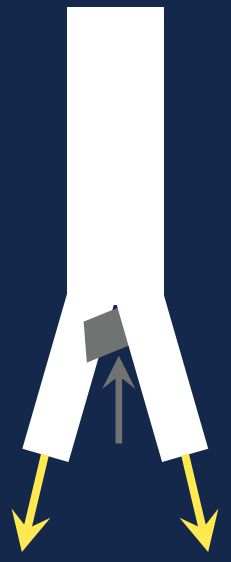




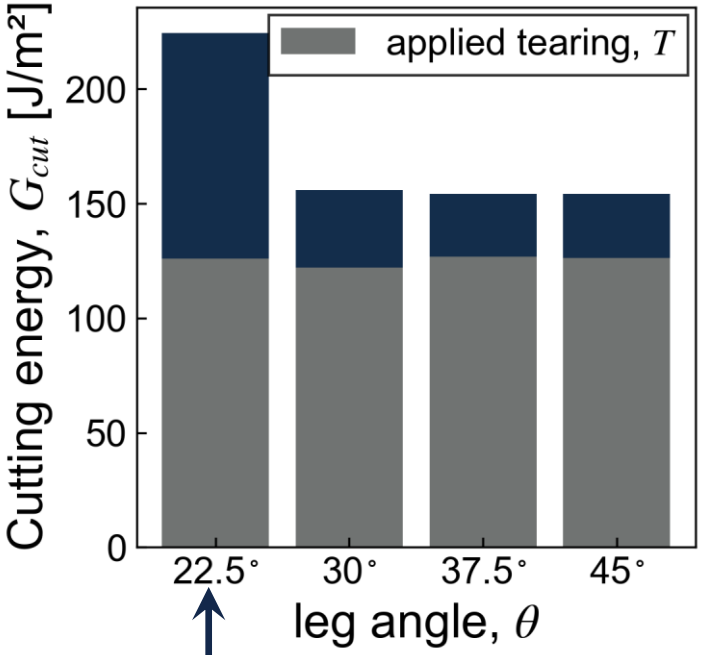


tear dependent cutting

Large deformation from applied far-field conditions also causes higher roughness.

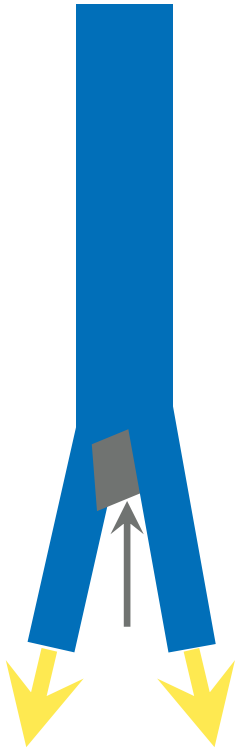


far-field boundary conditions



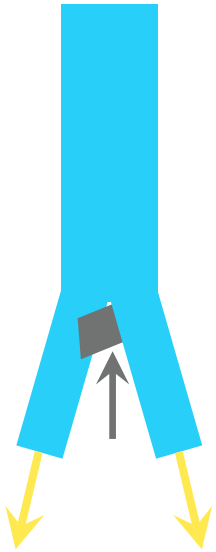
nonlinear G_{cut}

$\theta = 22.5°$

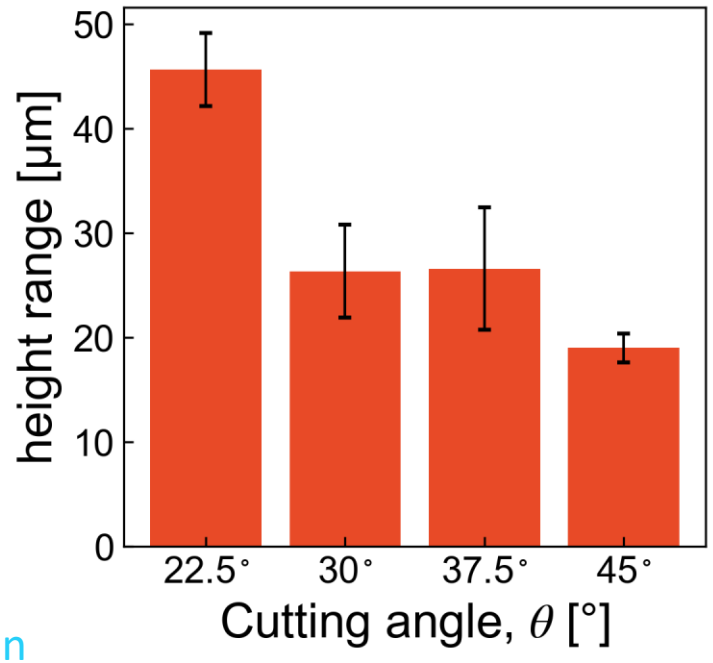


nonlinear deformation
$\lambda_{cut} \neq \lambda_{tear}$
$\lambda_{cut} \approx 1.2$
$\lambda_{tear} \approx 1.5$

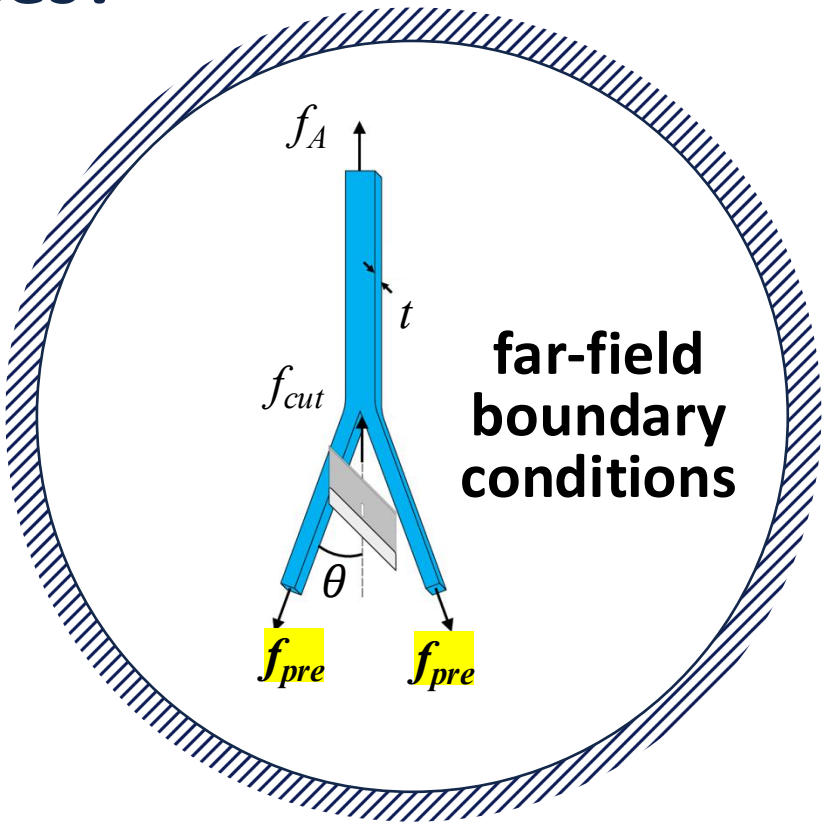
$\theta \geq 30°$

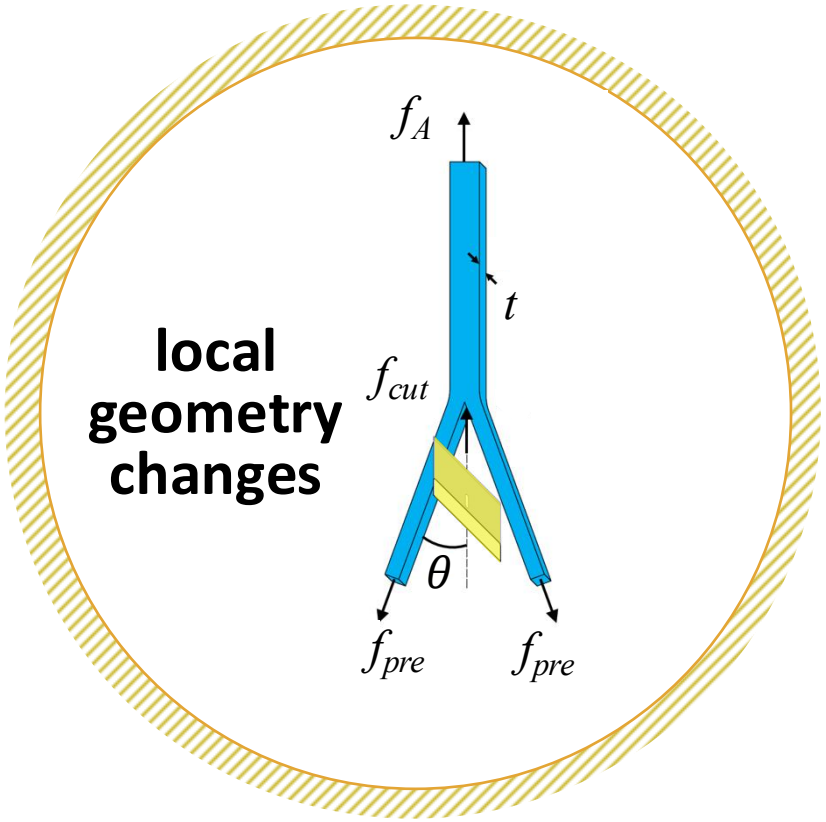


linear deformation
$\lambda_{cut} \approx \lambda_{tear} \approx 1 \pm 0.04$



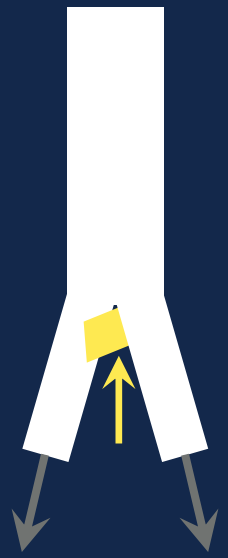
Is the apparent fracture energy reflected in the resulting cut surfaces?



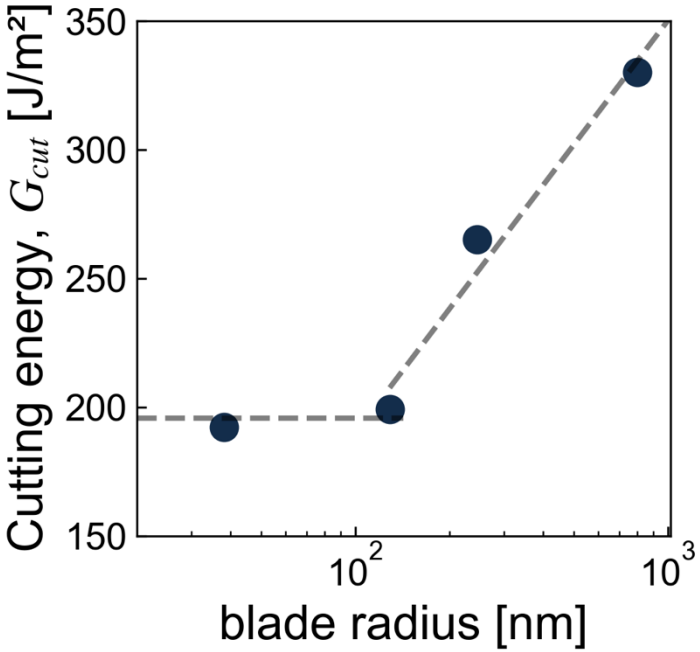


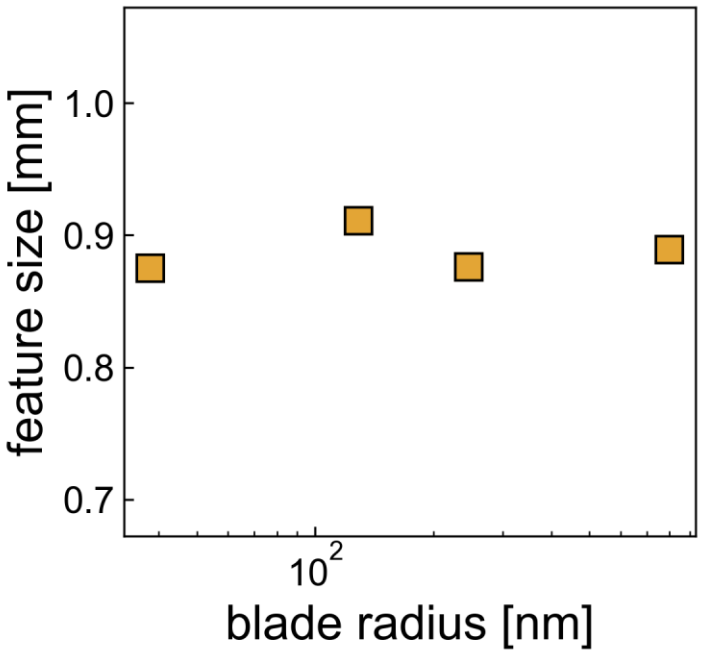
$E_{surface(failure\ zone)} \propto$ surface created

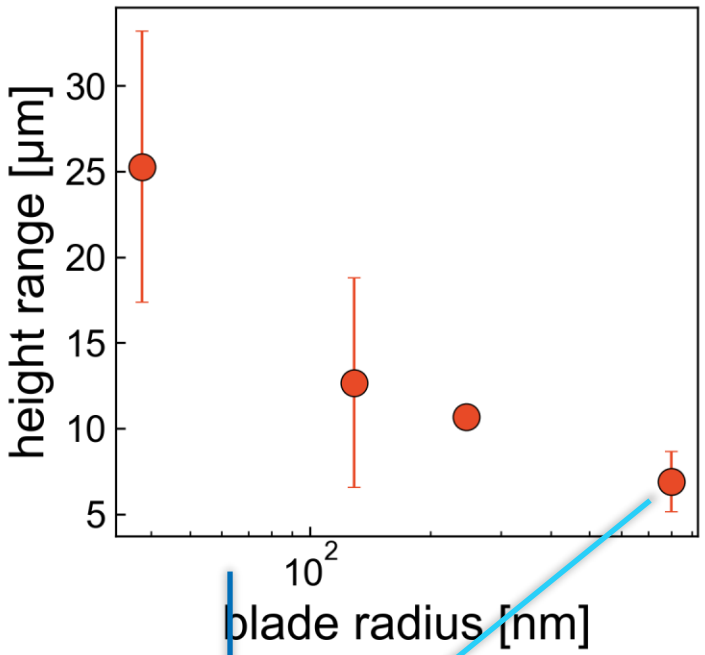
Increase in blade radius produces smoother surfaces - not proportional to fracture energy.

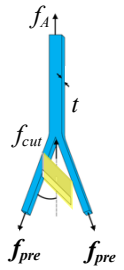


Local geometry changes

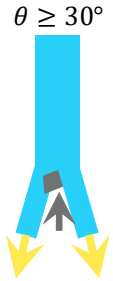






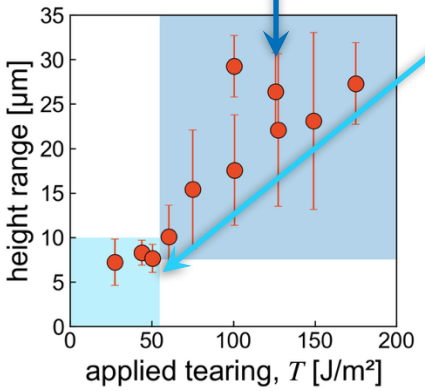


tear dependent cutting

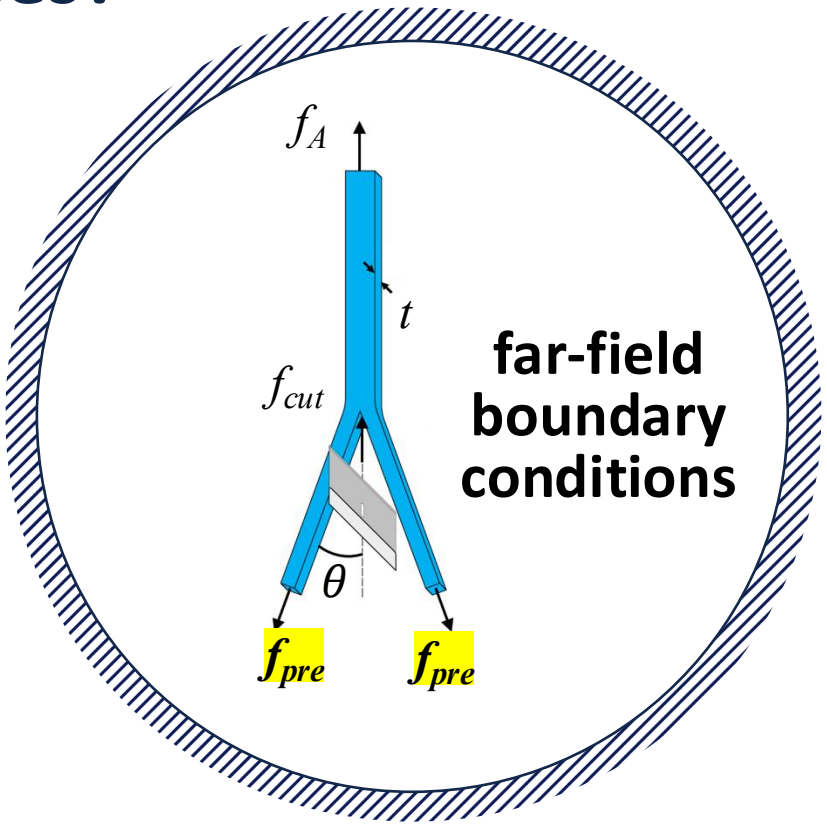


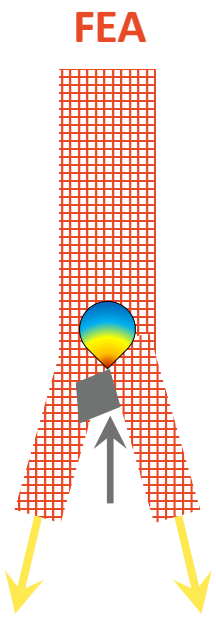
linear deformation
$\lambda_{cut} \approx \lambda_{tear} \approx 1 \pm 0.4$

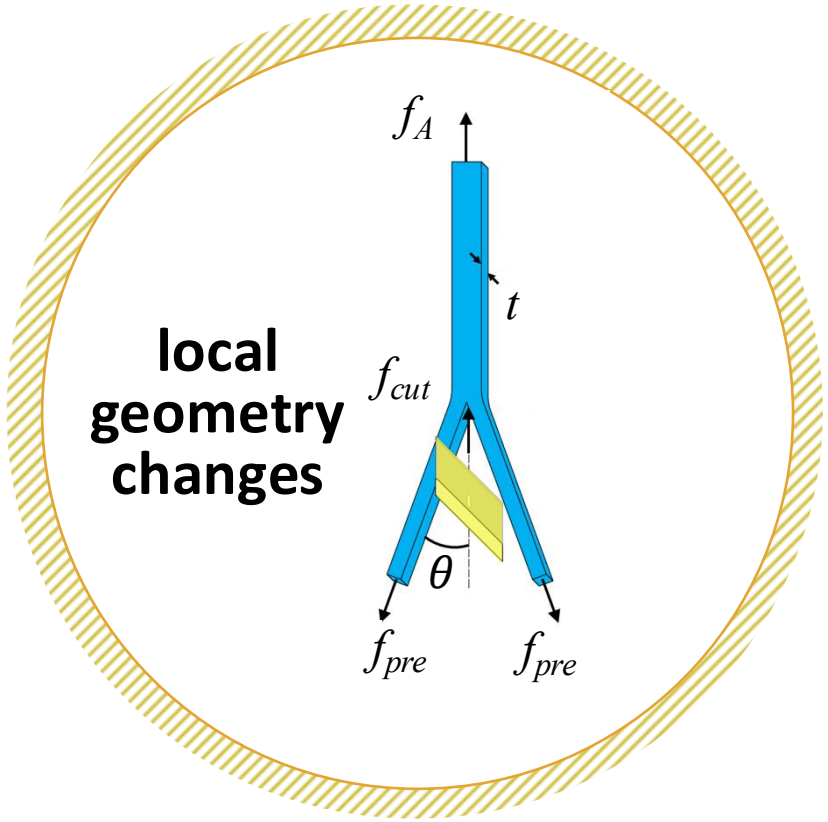
Contradictory to what was expected from previous results



Is the apparent fracture energy reflected in the resulting cut surfaces?



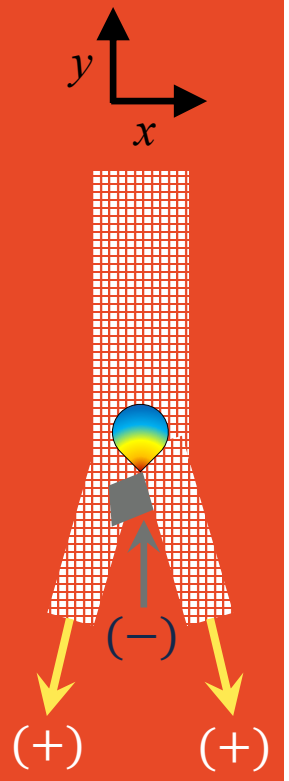




$E_{surface(failure\ zone)} \propto$ surface created

$E_{surface(failure\ zone)} \not\propto$ surface created

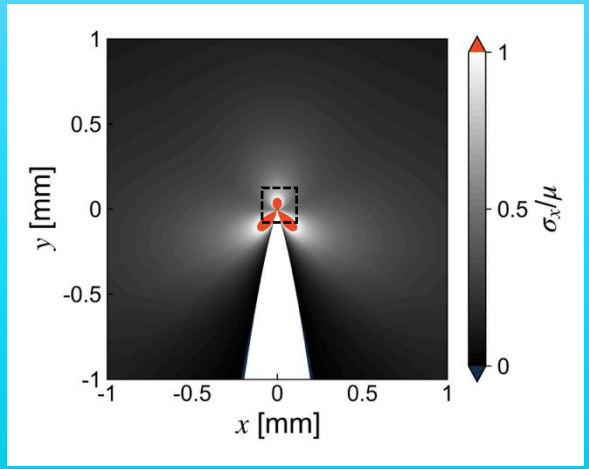
Dominant stress component dictates surface characteristics.

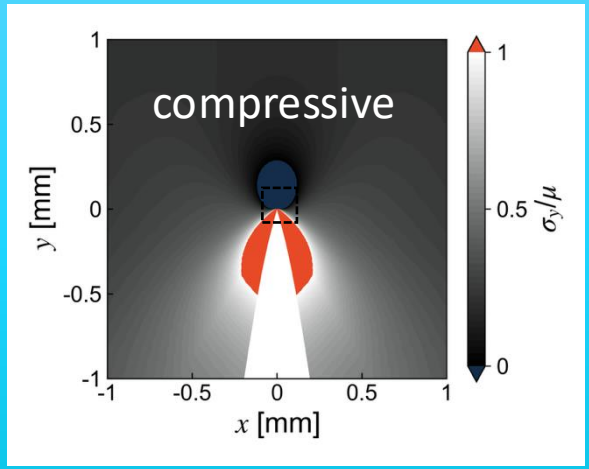


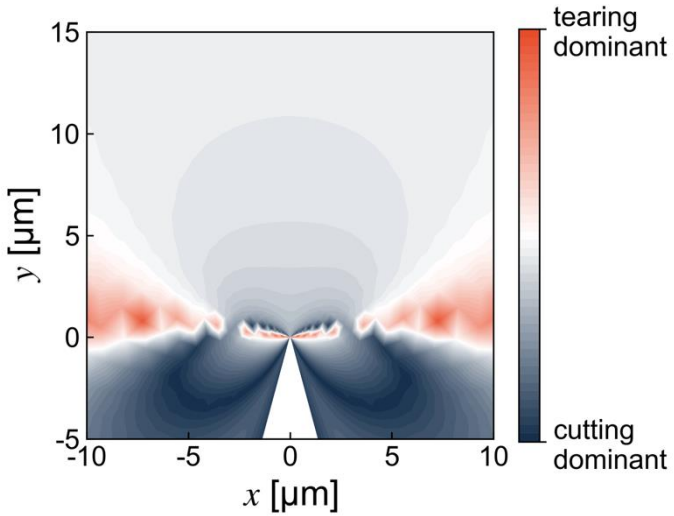
applied preload

cutting force from blade

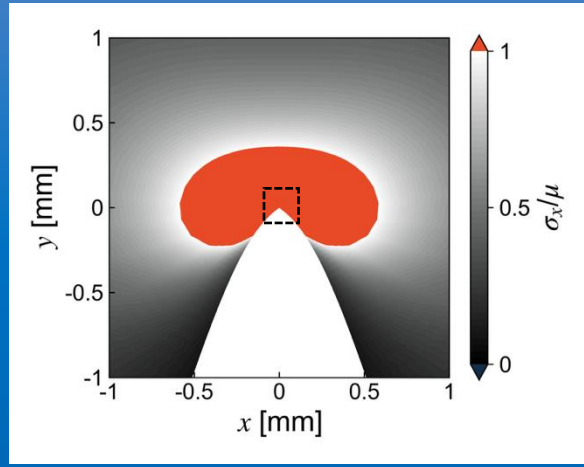
representative **'pure' cutting** simulation

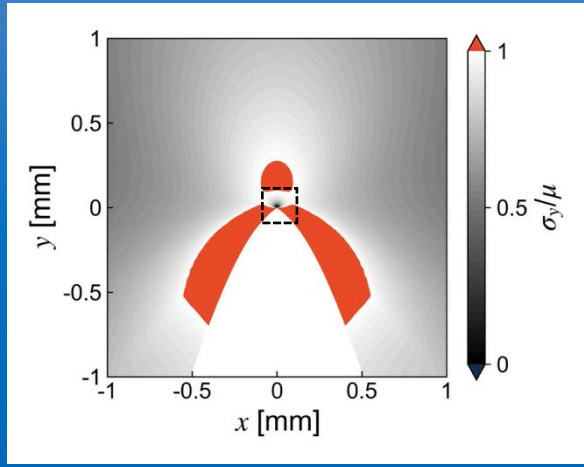


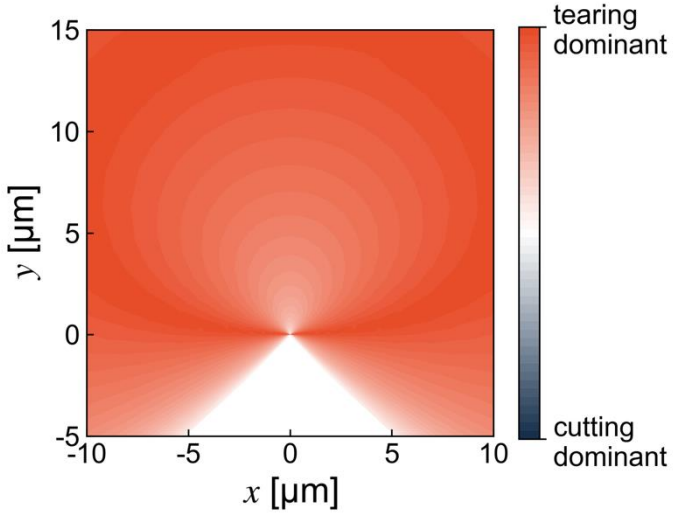


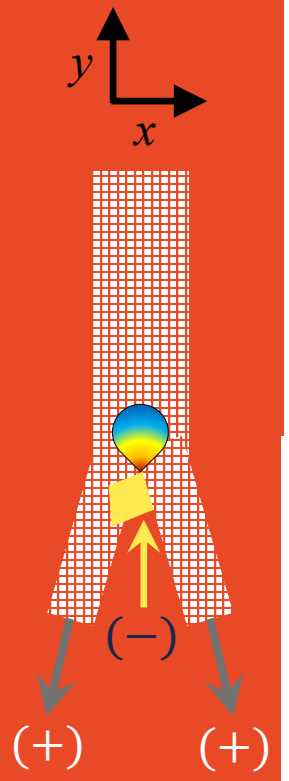


tear-dependent cutting simulation

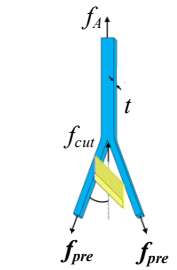




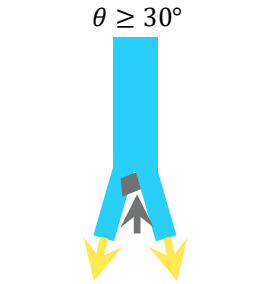




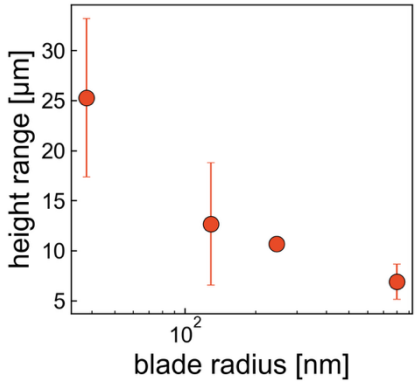
Local compression ahead of the blade dictates surface roughness.

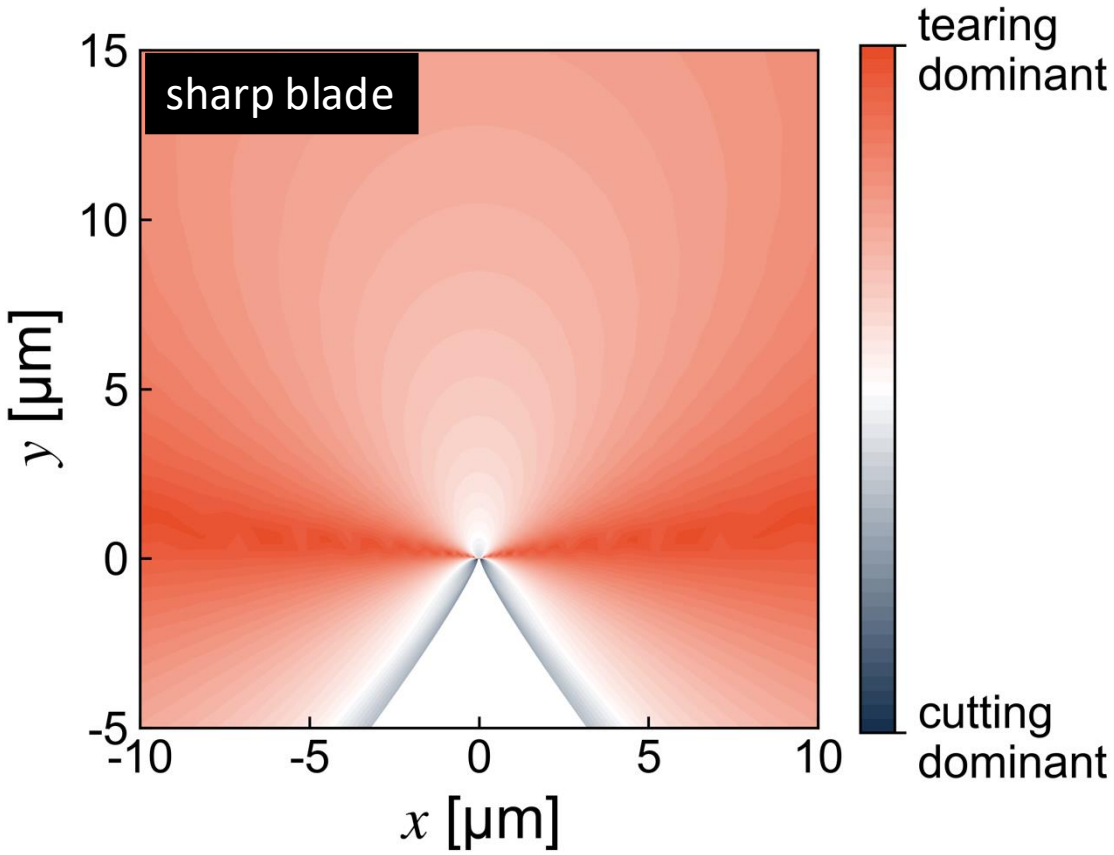


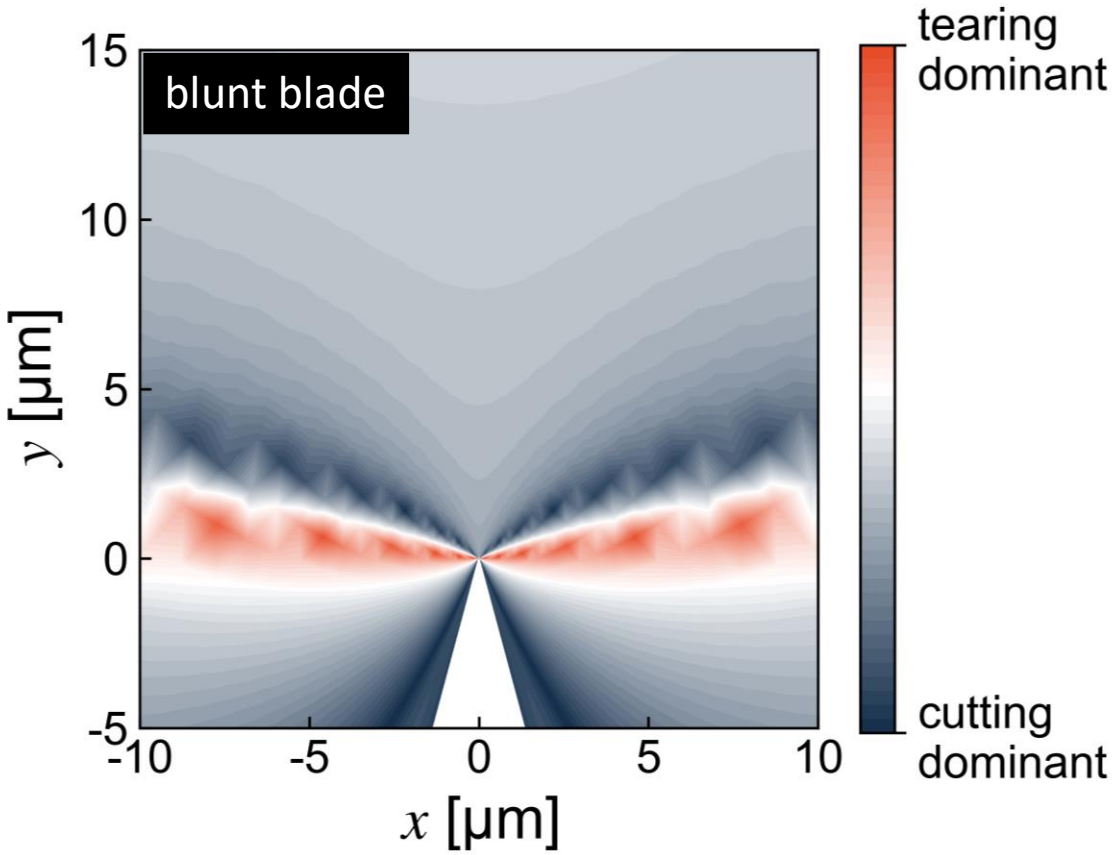
tear dependent cutting



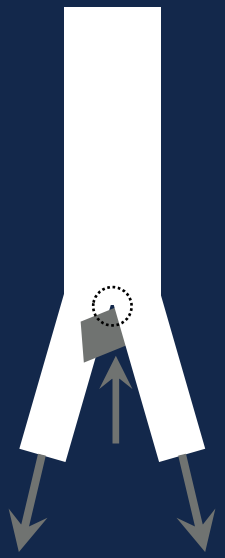
linear deformation
$\lambda_{cut} \approx \lambda_{tear} \approx 1 \pm 0.4$

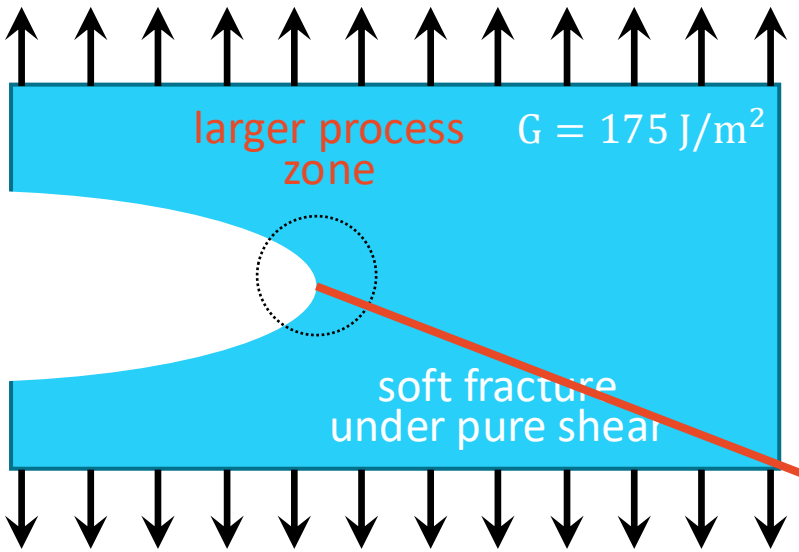


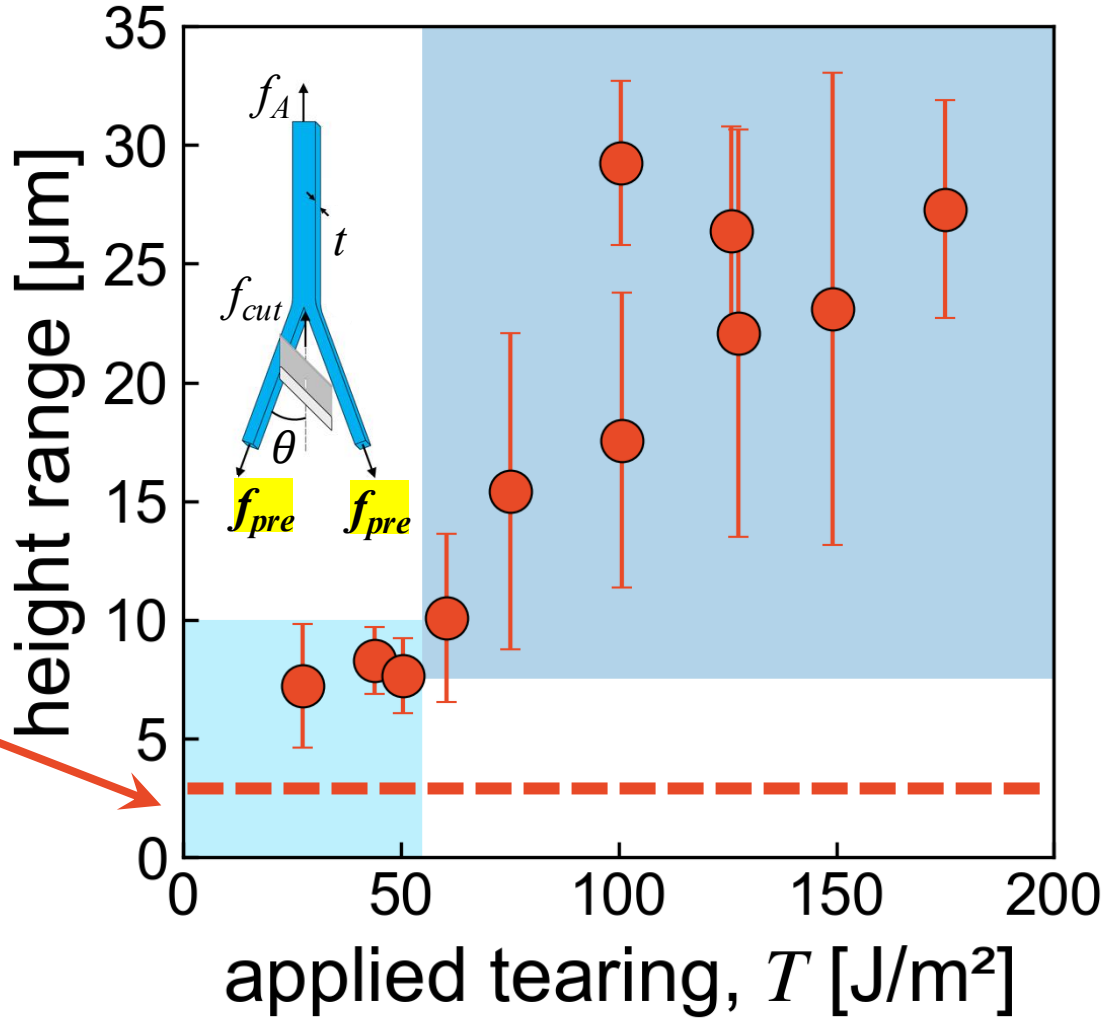




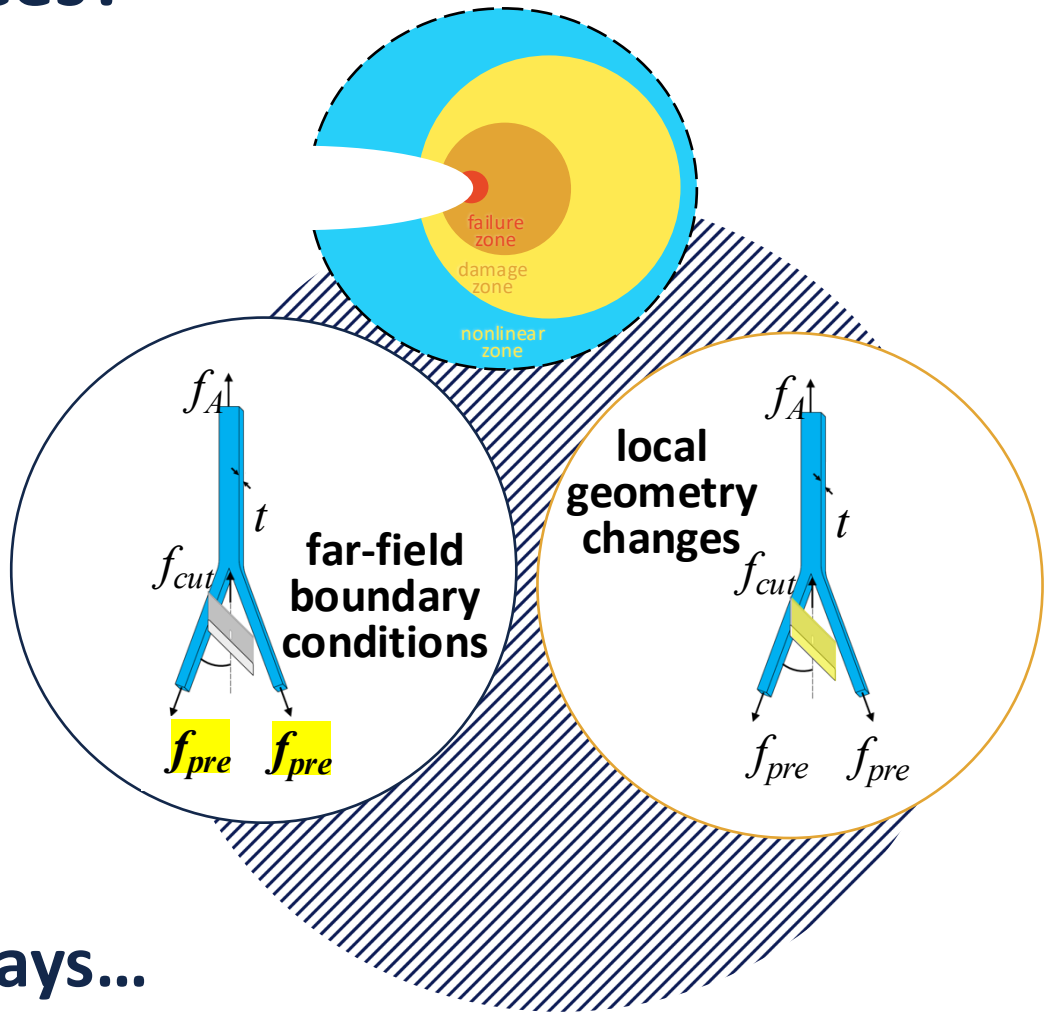
Except in pure shear, where the far-field stresses are higher at the local crack tip.



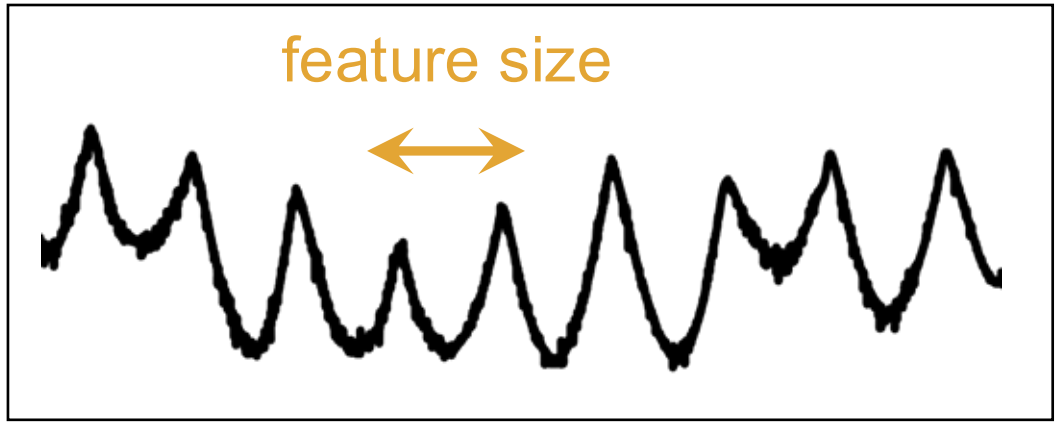




Is the apparent fracture energy reflected in the resulting cut surfaces?



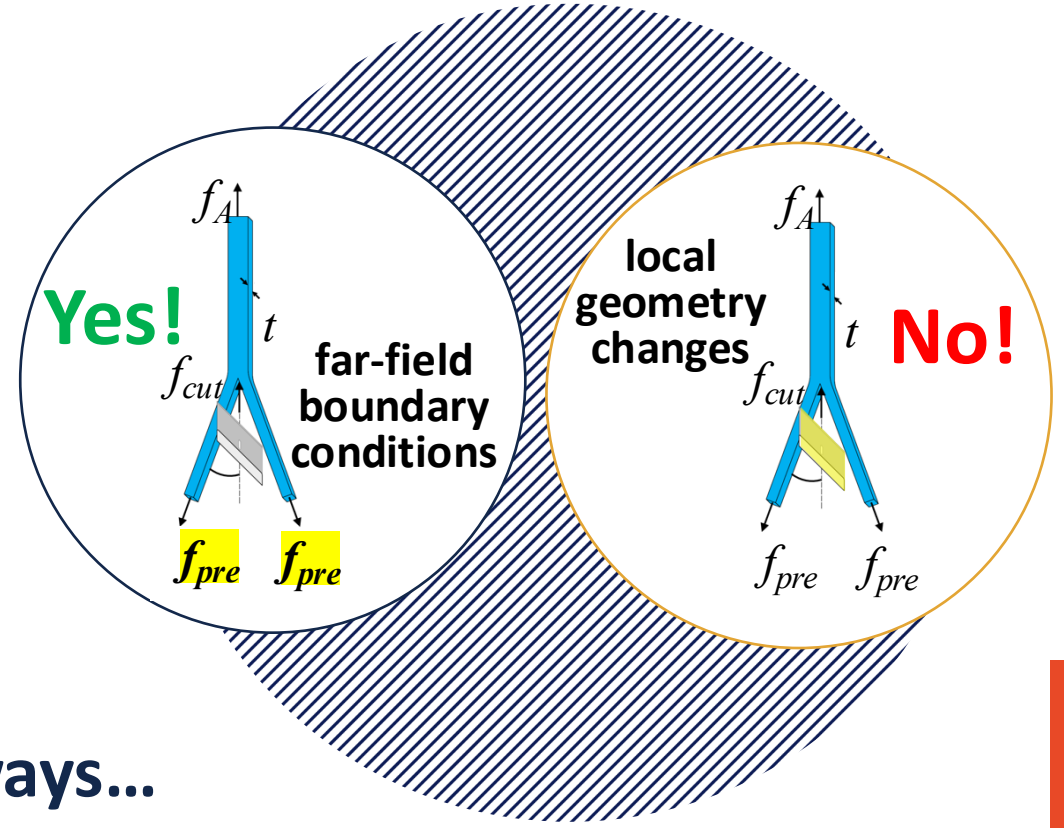
Periodicity is consistent across all boundary conditions.

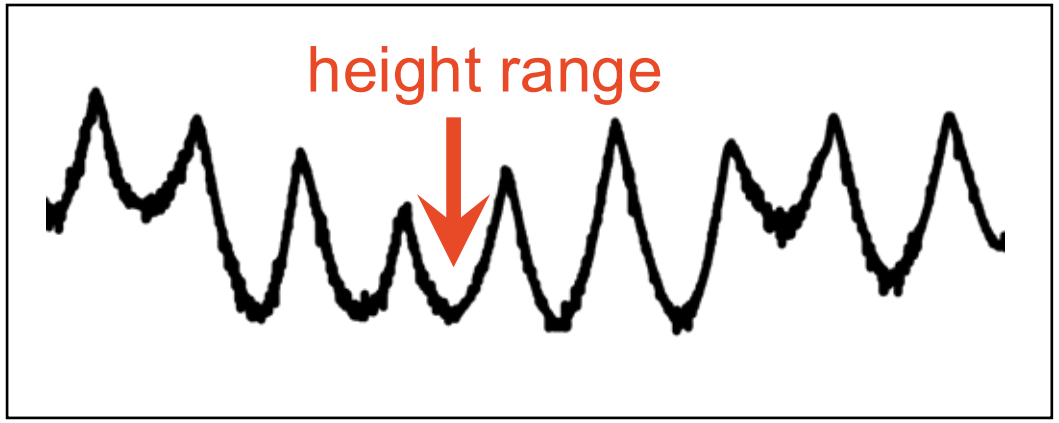


Takeaways…

Is the apparent fracture energy reflected in the resulting cut surfaces?

Higher <u>apparent</u> G_{cut} can produce rougher surfaces.





Takeaways…

Stress state ahead of the crack tip dictates surface roughness.

THANK YOU!

ILLINOIS

NSF

Award number:
2219787